REALIZING A PROFINITE SPACE AS THE CONNECTED COMPONENTS OF AN IND-ZARISKI COVER

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file proetale.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the presentation of the profinite space as a limit of finite discrete spaces, the construction of the ind-Zariski quotients realizing the closed subsets of the components, and the passage to the colimit with the identification of the components.

Editorial note. The label lemma-construct-profinite names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source proetale.tex –label lemma-construct-profinite –start 1097 –end 1139 –local-id 3593 returned RESULT=OK (L3593 free, occupancy FREE, public interval clear). Candidate taken from sources/stacks-proetale/candidates-halfA.json.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Description of a profinite space as a limit of finite discrete spaces. Topology, the lemma describing a profinite space as the limit of an inverse system of finite discrete spaces. Supplies the presentation of the profinite space T as a limit of finite discrete spaces T_i over a directed set.
- Construction of an ind-Zariski ring map from a closed subset of the components. Same chapter, the lemma constructing an ind-Zariski quotient corresponding to a closed subset of the connected components. Supplies, for each i, the surjection A_i -> B_i which is ind-Zariski and realizes the closed subset Z_i.
- Products of rings and local isomorphisms. Same chapter, the lemma that a local isomorphism is fully faithful and the lemma producing ind-Zariski maps between the pieces. Supply the description of X x T_i as the spectrum of the product of copies of A and the transition maps B_{i'} -> B_i.
- Identification of the connected components of the limit. Same chapter, the lemma identifying the connected components of the limit with the limit of the components. Supplies the conclusion that T is the space of connected components of Y.

Source reference. Original reference 097C: sources/stacks-proetale/pinned-original-source.tex, source lines 1081–1081. Prerequisite pointer invoked at proof line(s) 1118; it names a standard fact of the same chapter that the counted argument

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uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 096T: sources/stacks-proetale/pinned-original-source.tex, source lines 685–685. Prerequisite pointer invoked at proof line(s) 1126; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 096L: sources/stacks-proetale/pinned-original-source.tex, source lines 603–603. Prerequisite pointer invoked at proof line(s) 1125; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 00DS: sources/stacks-proetale/pinned-original-source.tex, source lines 434–434. Prerequisite pointer invoked at proof line(s) 1137; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 08ZY: sources/stacks-chow/tags.txt, source lines 11391–11391. Prerequisite pointer invoked at proof line(s) 1115; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let A be a ring and let $X = \mathrm{Spec}(A)$. Let T be a profinite space and let $T \to \pi_0(X)$ be a continuous map. There exists an ind-Zariski ring map $A \to B$ such that with $Y = \mathrm{Spec}(B)$ the diagram*

$$\begin{array}{ccc} Y & \longrightarrow & \pi_0(Y) \\ \downarrow & & \downarrow \\ X & \longrightarrow & \pi_0(X) \end{array}$$

is cartesian in the category of topological spaces and such that $\pi_0(Y) = T$ as spaces over $\pi_0(X)$.

Proof. Namely, write $T = \lim T_i$ as the limit of an inverse system finite discrete spaces over a directed set (see Topology, Lemma 08ZY). For each i let $Z_i = \mathrm{Im}(T \to \pi_0(X) \times T_i)$. This is a closed subset. Observe that $X \times T_i$ is the spectrum of $A_i = \prod_{t \in T_i} A$ and that $A \to A_i$ is a local isomorphism. By Lemma 097C we see that $Z_i \subset \pi_0(X \times T_i) = \pi_0(X) \times T_i$ corresponds to a surjection $A_i \to B_i$ which is ind-Zariski such that $\mathrm{Spec}(B_i) = X \times_{\pi_0(X)} Z_i$ as subsets of $X \times T_i$. The transition maps $T_i \to T_{i'}$ induce maps $Z_i \to Z_{i'}$ and $X \times_{\pi_0(X)} Z_i \to X \times_{\pi_0(X)} Z_{i'}$. Hence ring maps $B_{i'} \to B_i$ (Lemmas 096L and 096T). Set $B = \mathrm{colim}\, B_i$. Because $T = \lim Z_i$ we have $X \times_{\pi_0(X)} T = \lim X \times_{\pi_0(X)} Z_i$ and hence $Y = \mathrm{Spec}(B) = \lim \mathrm{Spec}(B_i)$ fits into the cartesian diagram

$$\begin{array}{ccc} Y & \longrightarrow & T \\ \downarrow & & \downarrow \\ X & \longrightarrow & \pi_0(X) \end{array}$$

of topological spaces. By Lemma 00DS we conclude that $T = \pi_0(Y)$. □